

BOOK REVIEWS | DECEMBER 01 1961

The Principles of Nuclear Magnetism

A. Abragam; L. C. Hebel



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Book Reviews

Modern Physics Buildings—Design and Function. R. RONALD PALMER AND WILLIAM MAXWELL RICE. Pp. 324+x. Reinhold Publishing Corporation, New York, 1961. Price \$13.50.

Anyone who has been intimately concerned with planning a physics laboratory for college or university will surely agree that this book will be extremely valuable to future planners. The decisions to be made in such an undertaking are innumerable, seldom have unique answers, and are mostly irrevocable. The planners strive for adaptability to growth in their design, but they must make many final commitments.

A sample of the general and specific questions which must be answered are the following:

What compromises shall be made among securing maximum space, maximum utility and adaptability, pleasing design, and finish?

What are the important considerations in laying out a library, shops, stockrooms, and other services?

What size and arrangement of an elementary physics laboratory are best?

Should direct current and special electrical services be distributed from central sources?

What alternating current power supply and distribution capacity are needed for anticipated simultaneous demands in experimental laboratories?

What heat load on air conditioning equipment will arise from experimental apparatus?

What, after all, is the best slope for the seats in the demonstration lecture room?

What is the most satisfactory kind of chalkboard?

The authors of the book do not answer these and other similar questions. What they do is present the accumulated experience of many attempts to solve such problems and thoughtfully discuss the basis for choices. Through examples and discussion they provide a great wealth of planning information.

Chapter I, entitled "Space requirements of physics," is especially to be recommended to architects who are approaching the design of a physics building for the first time. Chapter II outlines the steps in programming a building. Chapter III deals with the building as a whole, treating concisely such topics as costs, site planning, general features of construction and building components, acoustics and lighting, distribution of services, and equipment. Chapter IV is a collection of floor plans and photographs of 33 completed or projected buildings: 15 physics buildings in colleges or universities, 10 science buildings partly devoted to physics, 4 industrial laboratories, and 4 buildings in the design stage. Plans, photographs, some design, and cost data are given.

Chapters V–XI deal successively with the functional components of a physics building: lecture rooms, class and seminar rooms, teaching laboratories, research laboratories, shops and auxiliary rooms, offices and conference rooms, and library. Here is real meat for the departmental building committee confronted with the task of

transmitting to the architect the myriad of major and minor decisions which will determine the satisfaction that generations of students and professors will find in the use of the building. As noted above the planners will not find in the book unique or even optimum answers. They will find criteria on which to base decisions, and numerous alternatives which have been used, with a discussion of their advantages and shortcomings.

Chapter XII discusses laboratories and classrooms for physics in high school buildings. It contains descriptions of a number of innovations which may be useful in college laboratories.

The book contains many specific references to sources of additional information. An appendix lists these references as well as an excellent bibliography of pertinent general references.

The book contains more than 450 reproductions of photographs and line drawings (several are sometimes grouped in a single figure) which add enormously to its usefulness. Typography and reproduction are excellent.

The book is the report of the Project on the Design of Physics Buildings, conducted jointly by the AAPT and the AIP under a grant from the Educational Facilities Laboratories, Inc., an agency established by the Ford Foundation. Professor Palmer was the Director of the project from early 1959 through the completion of the book. Mr. Rice is an architect experienced in the design of scientific laboratories. All concerned are to be complimented on the completion of the project and of this much needed and extremely useful book.

G. M. ALMY
University of Illinois

The Principles of Nuclear Magnetism. A. ABRAGAM. Pp. 589. Clarendon Press, Oxford, England, 1961. Price \$13.44.

Since the discovery of nuclear magnetic resonance in 1945, the study of nuclear magnetism has produced an extensive literature through its application to chemistry and to solid state physics. With great success *The Principles of Nuclear Magnetism* draws together this varied literature, treating its many topics from a unified and up-to-date point of view. In addition, the author's spirited presentation emphasizes that nuclear magnetism, even apart from applications, is one of the most fascinating fields of modern physics.

The author gives a rigorous presentation of the theory behind nuclear resonance experiments and presents only enough experimental data to show where the theory works well, and where it is inadequate. The basic ideas underlying magnetic resonance—motion of free spins, Bloch equations, steady-state absorption and induction, and transient effects including spin echoes and fast passage—are treated from several points of view, accompanied by excellent pictures; very effective use is made of the rotating coordinate system in the discussion. The broadening and fine structure of resonance lines in solids due to di-

polar coupling and quadrupole interactions are thoroughly covered, with examples for both steady-state and transient types of measurements. However, while the treatment of multiplet structure and chemical shifts in liquids covers the fundamental ideas, a reader primarily interested in high-resolution nuclear spectroscopy will need to draw on more specialized supplementary material.

The best sections, which make up half the book, deal with relaxation phenomena—thermal relaxation in liquids and gases, thermal relaxation and dynamic polarization in solids, linewidth in the presence of motion, and the discussions of spin temperature as well as effects of strong radio-frequency fields (saturation). By using the theories of Kubo and Tomida, Bloch, and Redfield to supplement older treatments, the author achieves a clear and unified discussion of many relaxation mechanisms applied to a wide variety of situations. The data presented, while sparse, clearly indicates where the theory stands.

In such a treatment there is not sufficient room to discuss current experimental techniques in detail; a brief 22-page summary of the principle methods serves as a review. There is also no room to discuss applications to nuclear physics. However, the book will be very useful to those who wish to know in detail what nuclear magnetism reveals about a wide variety of substances, and it will help the worker in the field in the exposition of his research. Surely, *The Principles of Nuclear Magnetism* will become the standard text for graduate students entering this or any of the closely related fields of research.

L. C. HEBEL
Bell Telephone Laboratories, Inc.

Principles of Fluid Mechanics. RICHARD A. KENYON. The Ronald Press Company, New York, 1960.

Kenyon's book has many interesting features. In the first place, it contains some material not usually found in an elementary book on fluid mechanics; for example, shock waves and Newtonian fluids. The diagrams are clear; the problems look interesting. I myself would have preferred to have seen more material on common phenomena like surface waves and capillarity, as well as on the more unusual supersonic flows—not to mention thermal properties of subsonic flows.

The book claims to be "based solidly on a background of thermodynamics" and to differ "from other elementary texts in the depth and rigor with which fundamental principles are presented." It is in exactly this regard, however, that physicists may have some hesitation in using it. I am not referring primarily to statements such as "The student will recall from his elementary physics . . . $a = [g_c(dP/d\rho)_s]^{\frac{1}{2}}$." To be sure, having never seen that particular formula before, I could not recall it. Earlier, one is introduced to the dimensional constant $g_c (= 32.174)$ in the expression $F = ma/g_c$. It arises because of the engineering usage (simultaneous) of a pound-mass unit and of a pound-force unit. It does emphasize, however, one minor

drawback of the book, i.e., too many references to material presumably to be found elsewhere. A student might be puzzled also about "the hydrostatic equation for both compressible and incompressible fluids." Apparently the relationship between pressure and height in a fluid is so glorified.

Much more significantly, however, a critical student would undoubtedly encounter difficulties with remarks such as "Energy equation is derived from the principle of the conservation of energy," and "The general three-dimensional equation of continuity is derived from the concept of conservation of mass." Are not these tautological? One is not always clear also about the physical meaning of such statements as "Weight may be defined as the radially inward force exerted by a given mass owing to the local acceleration of gravity," or "Temperature is a thermodynamic fluid property that measures the internal energy level or molecular motion of a fluid or a substance." Moreover, the polytropic equation is introduced without any definition of polytropic as referring to a constant specific heat. The equation $PV = TdS - PdV$ is given without any reference to the process being necessarily reversible. A perfect gas is described simply as one obeying the equation $PV = RT$ without any additional criteria as to the internal energy being a function of absolute temperature alone and of the specific heat itself being constant.

RAYMOND J. SEEGER
National Science Foundation

Quality Science for Secondary Schools. Pp. 210, National Science Teachers Association, Washington, D. C., 1960. Price \$3.00.

This book, to quote from the Foreword, consists of "suggestions for improving the science program and specific courses in junior and senior high schools, with special reference to the continuous science program, kindergarten through twelfth grade." It was prepared by a committee of the National Science Teachers Association and is designed to aid secondary school principals and other school administrators in the redesign of their science curricula.

There is a lengthy discussion in the first chapter (by Donald Decker, President, 1959-60, of the NSTA) of something called the "quality of concepts in science" which this reviewer still doesn't really understand. Presumably it has something to do with the title of the book but it rapidly gets lost by the wayside in the remaining chapters. I suspect the term is just educational jargon for the degree of understanding of a scientific phenomenon.

The book suffers from an overabundance of authors and contributors (eighteen) writing independently with an excessive amount of overlap. School administrators deserve a little consideration. They are busy people and should not have to read 210 pages of text to glean 40 pages worth of useful information about trends in school science. One is reminded of the complaint of science teachers required to amass large numbers of credits in education at the expense of subject matter courses. They admit the educa-